

WORKED KEY

Full Name: _____

Period: _____

This practice is built exactly like the real test — same questions, same order, different numbers. Work every problem, then check the worked key. Anything you miss is exactly what to study.

1. Graph the linear equation: $y = 4x - 2$

Answer: start (0, -2); rise 4 run 1 → (1, 2), (2, 6)

2. Graph the linear equation: $y = -(3/5)x + 1$

Answer: start (0, 1); fall 3 run 5 → (5, -2)

3. Graph the linear equation: $x = 3$

Answer: vertical line through (3, 0)

4. Write the equation, in slope-intercept form, of the line with slope 5 and y-intercept -8.

Answer: $y = 5x - 8$

5. Find the slope of the line passing through (-1, -5) and (4, 10).

Answer: $m = (10 - (-5))/(4 - (-1)) = 15/5 = 3$

6. Find the slope of the line passing through (-5, -6) and (2, -6).

Answer: $m = (-6 - (-6))/(2 - (-5)) = 0/7 = 0$ — equal y-values make a horizontal line

7. A line has a slope of 3 and passes through (2, y) and (5, 16). Find the missing coordinate.

Answer: $(16 - y)/(5 - 2) = 3 \rightarrow 16 - y = 9 \rightarrow y = 7$

Algebra 1 – Module 4 Practice Test •

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8. Write the equation $5x - 2y = 12$ in slope-intercept form, then identify the slope and y-intercept.

Answer: $-2y = -5x + 12 \rightarrow y = (5/2)x - 6$; $m = 5/2$, $b = -6$

9. For the line $y = (4/3)x - 5$, identify the slope and the y-intercept, and explain what each tells you about the graph.

Answer: $m = 4/3$ — the line rises 4 for every 3 right; $b = -5$ — it crosses the y-axis at $(0, -5)$

10. Does the table show a linear relationship? If so, give the constant rate of change.

x	y
0	-2
3	13
6	28
9	43

Answer: constant rate: $(13 - (-2))/(3 - 0) = 15/3 = 5$ per unit — yes, it is linear

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11. Error Analysis — To graph $y = -5x + 6$, Rex plotted his first point at the point $(6, 0)$. Identify his mistake.

Answer: $b = 6$ is the Y-intercept — it belongs at $(0, 6)$ on the y-axis, not at $(6, 0)$

12. The battery charge left on a delivery drone is modeled by $c = -6t + 42$, with c in percent and t in minutes of flight. Interpret the slope and the y-intercept, then find when the model reaches zero.

Answer: the slope means the battery loses 6% each minute; the 42 is the charge at takeoff; $c = 0 \rightarrow -6t + 42 = 0 \rightarrow t = 7$ minutes

13. Write the equation, in slope-intercept form, of the line through $(0, 7)$ and $(3, -5)$.

Answer: $m = (-5 - 7)/3 = -4$, $b = 7$ (given by $(0, 7)$) $\rightarrow y = -4x + 7$

Cumulative Review — ACT Style. Circle the best answer.

14. Which relation is a function?

- Ⓐ $\{(2, 4), (2, 5), (3, 6)\}$
 Ⓑ $\{(8, 1), (8, 2), (9, 3)\}$
 Ⓒ $\{(0, 3), (1, 4), (0, 5)\}$
 Ⓓ **$\{(4, 7), (5, 7), (6, 8)\}$**

Answer: repeated OUTPUTS are fine; only the fourth has no repeated input

15. Solve: $6(x + 3) = 48$

- Ⓔ **5**
 Ⓕ 6
 Ⓖ 8
 Ⓗ 11

Answer: $x + 3 = 8 \rightarrow x = 5$

16. Solve $A = lw$ for w .

- Ⓐ $w = Al$
 Ⓑ $w = l/A$
 Ⓒ $w = A - l$
 Ⓓ **$w = A/l$**

Answer: divide both sides by l